# Learning LaTeX Through a Simple Article

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## 1 Introduction

This article is designed to show how a complete LaTeX document looks. It demonstrates document classes, titles, sectioning, lists, and mathematical symbols.

### 1.1 Motivation

LaTeX is used for producing professional documents.

#### 1.1.1 Why Students Need It

Because it ensures formatting is consistent.

**Key Point** Learning LaTeX early saves time in research.

**Mini Note** Even small reports benefit from LaTeX.

## 2 Organizing Information

Information is often presented as lists. LaTeX supports both itemized and numbered lists.

### 2.1 Custom Itemize

- Normal bullet

! Urgent task

■ Completed work

NOTE Special note

## 2.2 Numbered List

1. First numbered point

! Custom exclamation

■ Black square instead of number

NOTE Replace number with text

## 2.3 Nested Example

1. Outer Level

   - Inner bullet

   Note: This is a note

   Wait! This is a warning

# 3 Mathematics in Action

By practicing this, you now cover all the essential commands from the tutorial.

## 3.1 Summation Example

$$S_n = \frac{1}{n} \sum_{k=1}^{n} \frac{k^2 + k + 1}{k^2 + 1}$$

## 3.2 Geometric Series

$$1 + r + r^2 + r^3 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$